

GWP #3

March 28, 2026

1 Executive Summary

Table 1: Model performance and backtest summary across all steps.

step	model	accuracy	roc_auc	total_return	max_drawdown	sharpe	hit_rate
1	cnn_gaf	0.584	0.521	0.548	-0.140	1.315	0.575
1	lstm	0.585	0.474	0.606	-0.192	1.417	0.585
1	mlp	0.580	0.495	0.526	-0.192	1.271	0.580
2a	cnn_gaf	0.545	0.504	0.927	-0.264	1.049	0.545
2a	lstm	0.543	0.506	1.098	-0.264	1.174	0.542
2a	mlp	0.536	0.511	0.567	-0.341	0.746	0.536
2b	cnn_gaf	0.556	0.522	0.051	-0.244	0.370	0.556
2b	lstm	0.536	0.485	0.005	-0.243	0.148	0.536
2b	mlp	0.546	0.503	0.090	-0.180	0.547	0.546
3b	cnn_gaf	0.546	0.449	0.919	-0.192	1.186	0.546
3b	lstm	0.546	0.530	0.996	-0.192	1.253	0.546
3b	mlp	0.530	0.513	0.182	-0.386	0.367	0.529
3c	cnn_gaf	0.604	0.431	0.052	-0.150	0.613	0.604
3c	lstm	0.604	0.483	0.065	-0.139	0.738	0.604
3c	mlp	0.622	0.459	0.070	-0.139	0.779	0.613

2 Step 1

2.1 Step 1a. Data description

Item	Value
Security	SPY
Data source	Yahoo Finance via yfinance (auto-adjusted)
Frequency	Daily
Sample	2018-02-15 – 2025-12-31
Observations	1980
Project constraint	≤ 2000 observations (satisfied)
Lookback window	32 observations
Label horizon	1 observation(s) ahead
Target definition	$y_t = \mathbb{I}[r_{t+1} > 0]$; class-1 rate=0.553
Feature set	ret, ret_mean_5, ret_mean_10, ret_mean_20, ret_std_5, ret_std_10, ret_std_20, rsi_14, macd, macd_signal, log_volume, log_volume_z20
Trading cost assumption	5.0 bps per unit turnover

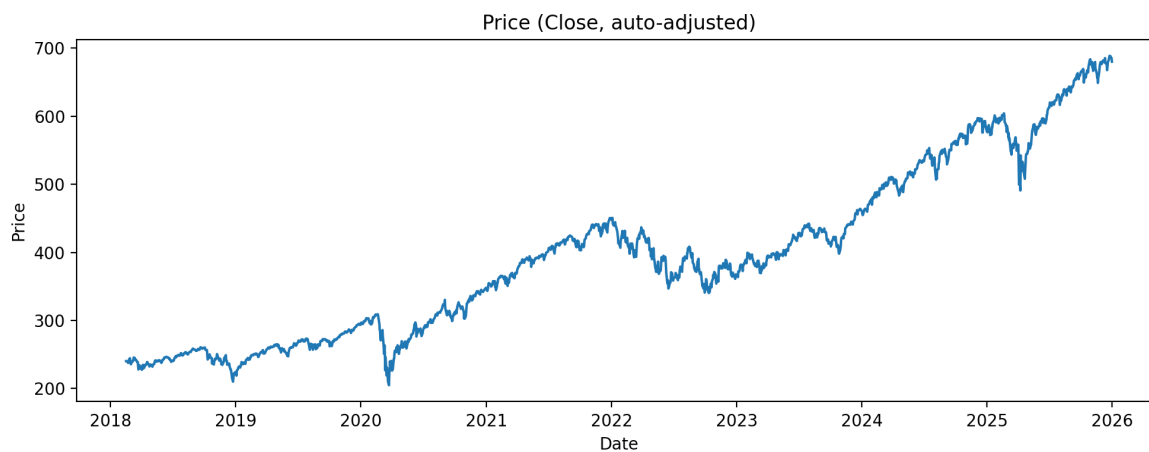


Figure 1: Auto-adjusted close price for the selected security.

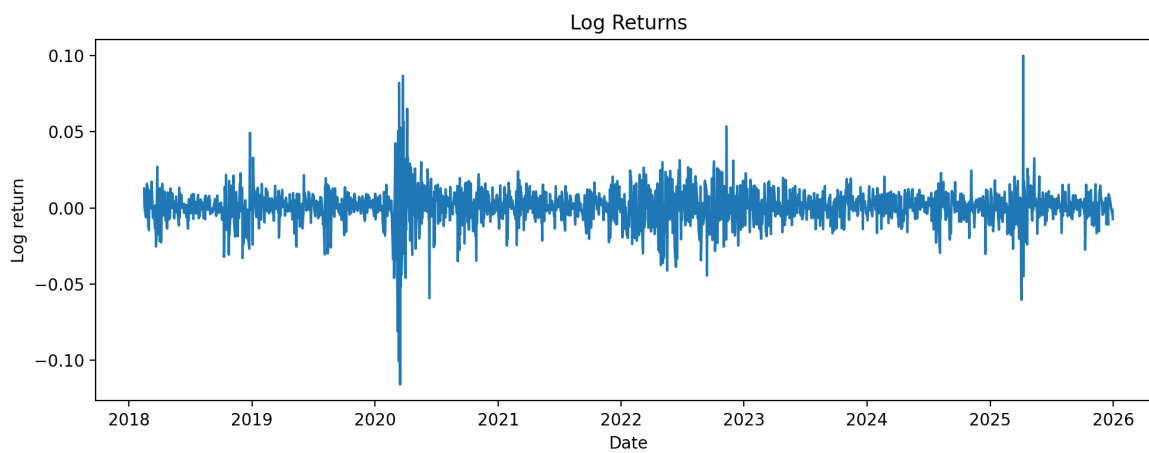


Figure 2: Log returns time series used for feature engineering and labeling.

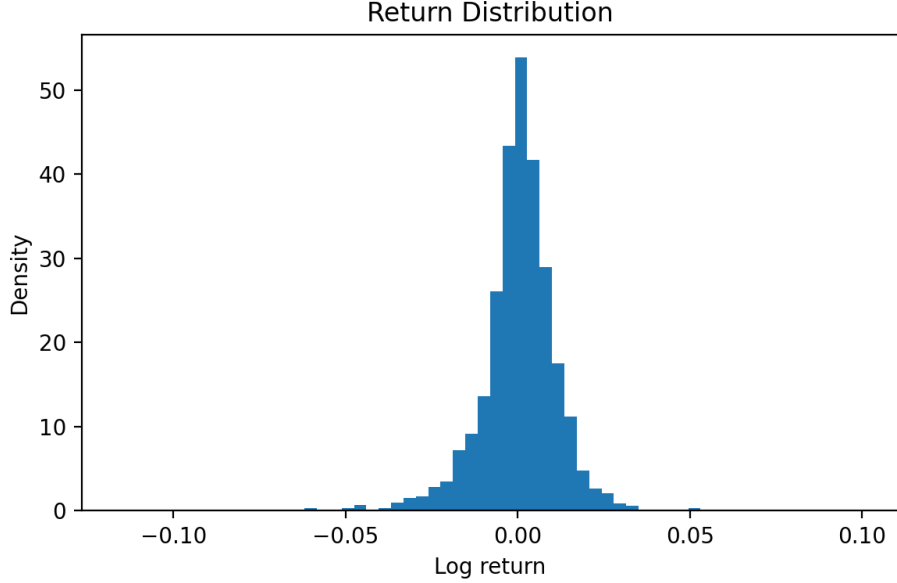


Figure 3: Empirical distribution of log returns.

2.2 Step 1b. Label construction and leakage (intentional)

We constructed a leaky modeling setup to show backtest overfitting due to information leakage. We standardize features using a **single scaler fit on the full dataset** (train+test) before splitting, so statistics from the future test period influence the training representation. For the CNN+GAF model, we also normalize returns using global min/max over the full sample when mapping windows into Gramian Angular Field images; this similarly contaminates the feature construction with future information. This leakage can inflate apparent predictive accuracy and trading performance because the model indirectly benefits from knowledge of the test distribution, a common pitfall in financial ML research and backtesting (López de Prado; Bailey et al.).

2.3 Step 1c. Single-split prediction results

We use a single chronological train/test split and train three deep learning classifiers:

- **MLP**: a feed-forward network on flattened lookback windows.
- **LSTM**: a single-layer LSTM with hidden size 64 followed by a small dense head (Hochreiter and Schmidhuber).
- **CNN on GAF**: a small 2-layer CNN trained on GAF images constructed from return windows (Wang and Oates) and a convolutional architecture (LeCun et al.).

All models use lookback length 32, AdamW optimization (lr=0.001, weight decay=0.0001), and early stopping on a chronological validation split inside the training window. Prediction performance on the held-out test period is summarized in Table 1.

2.3.1 Step 1c(1). MLP

2.3.2 Step 1c(2). LSTM

2.3.3 Step 1c(3). CNN on GAF

2.4 Step 1d. Backtests from single-split models

Trading rule: go long when $\hat{p}(\text{up}) \geq 0.5$, otherwise short; transaction costs are 5.0 bps per unit turnover.

Model	Acc	AUC	Sharpe	TotalRet	MaxDD	HitRate
mlp	0.580	0.495	1.271	0.526	-0.192	0.580
lstm	0.585	0.474	1.417	0.606	-0.192	0.585
cnn_gaf	0.584	0.521	1.315	0.548	-0.140	0.575

2.4.1 Step 1d(1). Strategy from MLP

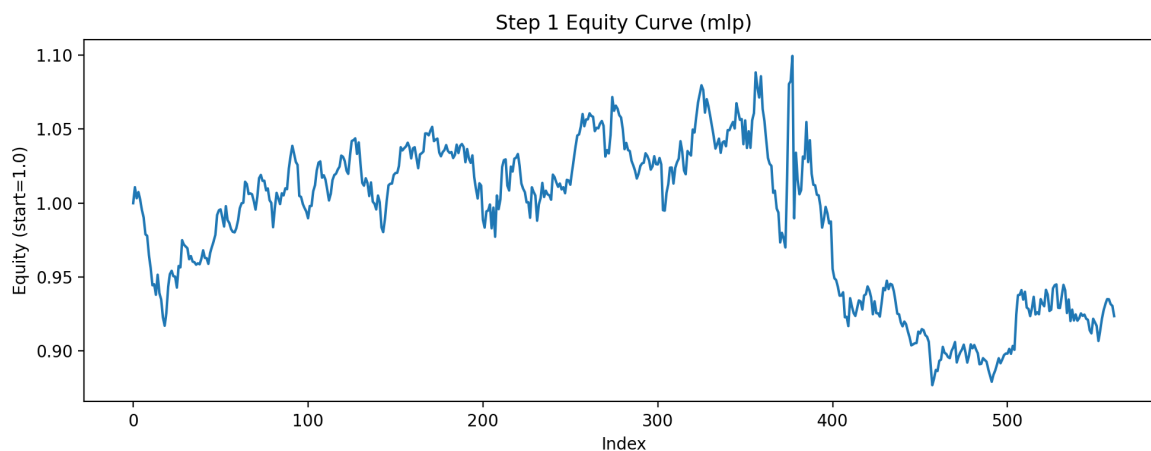


Figure 4: Step 1 backtest equity curve implied by the MLP.

2.4.2 Step 1d(2). Strategy from LSTM

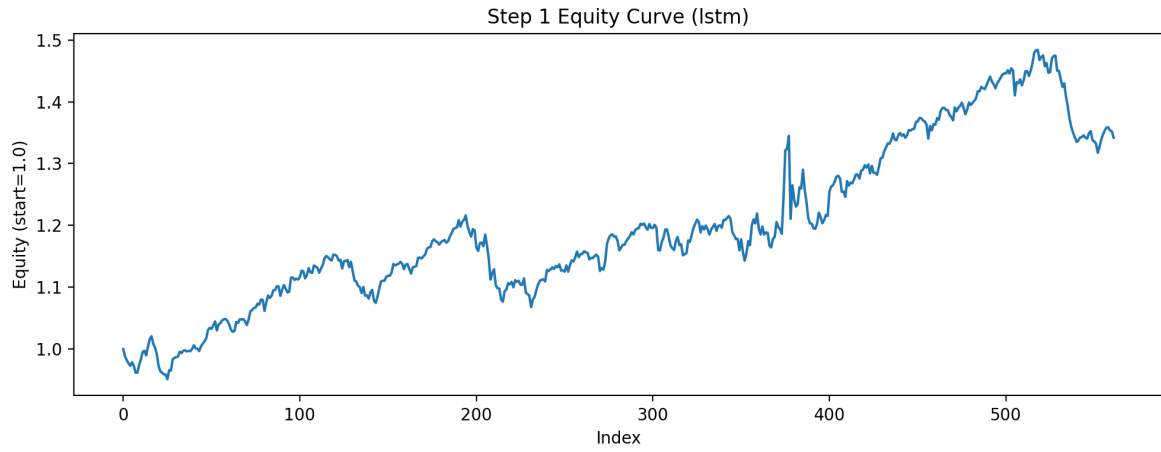


Figure 5: Step 1 backtest equity curve implied by the LSTM.

2.4.3 Step 1d(3). Strategy from CNN on GAF

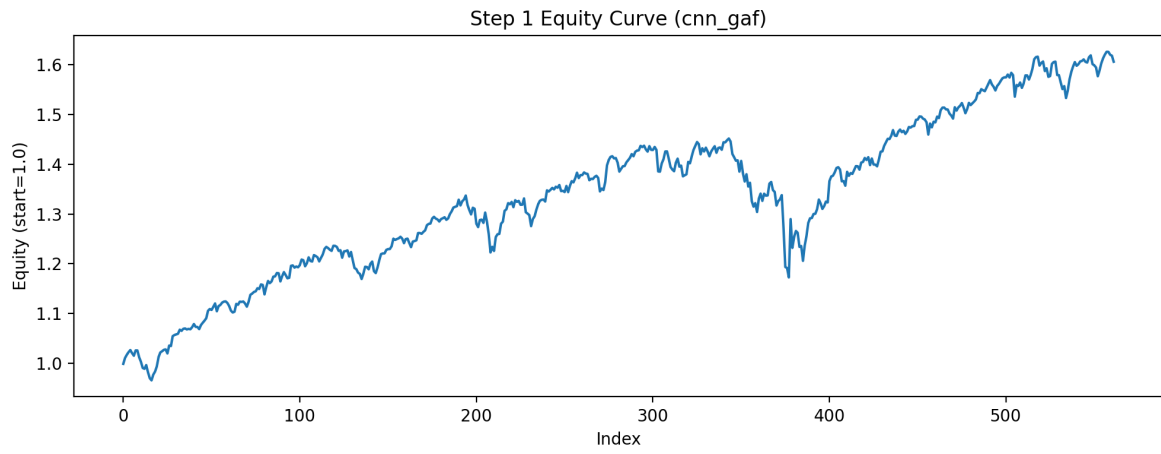


Figure 6: Step 1 backtest equity curve implied by the CNN on GAF features.

3 Step 2 (Walk-forward backtests with leakage present)

3.1 Step 2a. Non-anchored walk-forward with 500/500 splits

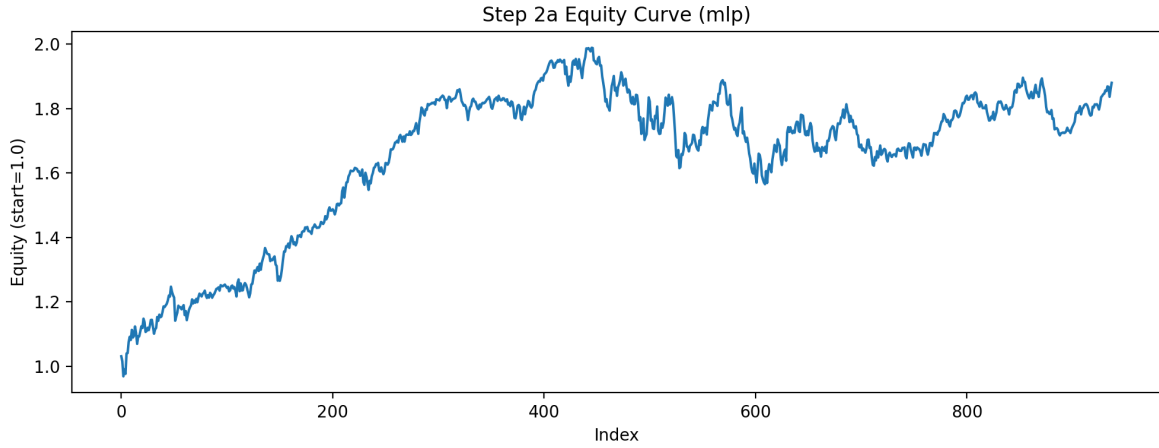


Figure 7: Step 2a aggregated walk-forward equity curve (MLP).

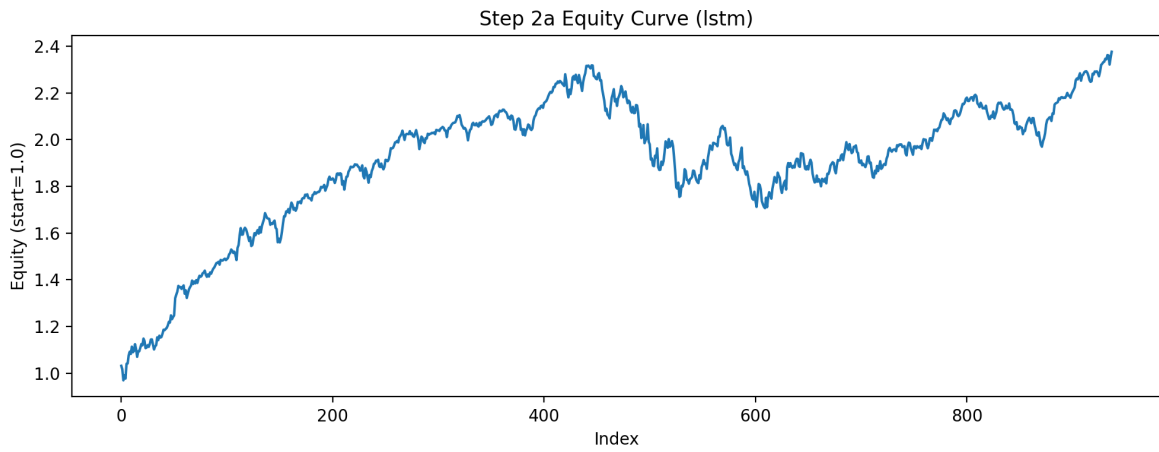


Figure 8: Step 2a aggregated walk-forward equity curve (LSTM).

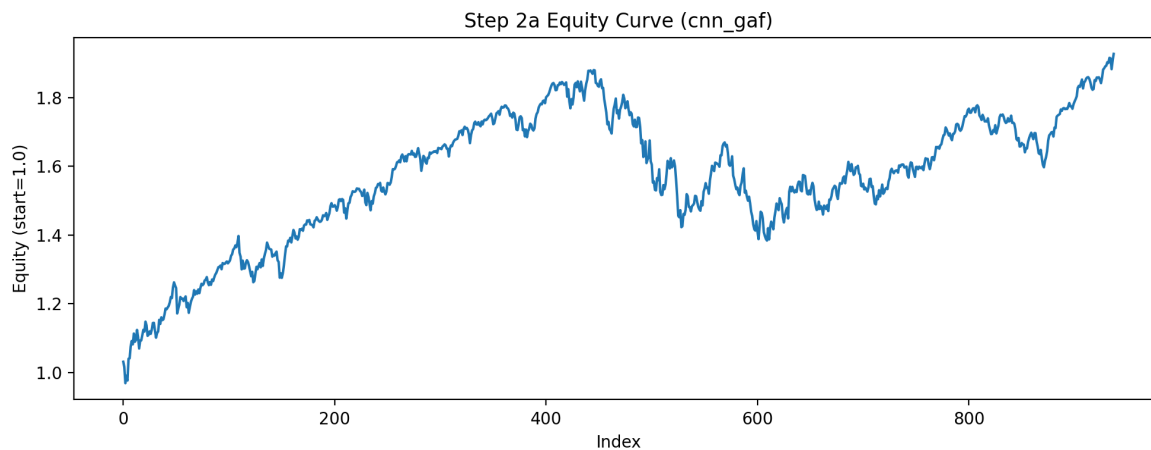


Figure 9: Step 2a aggregated walk-forward equity curve (CNN on GAF).

3.2 Step 2b. Non-anchored walk-forward with 500/100 splits

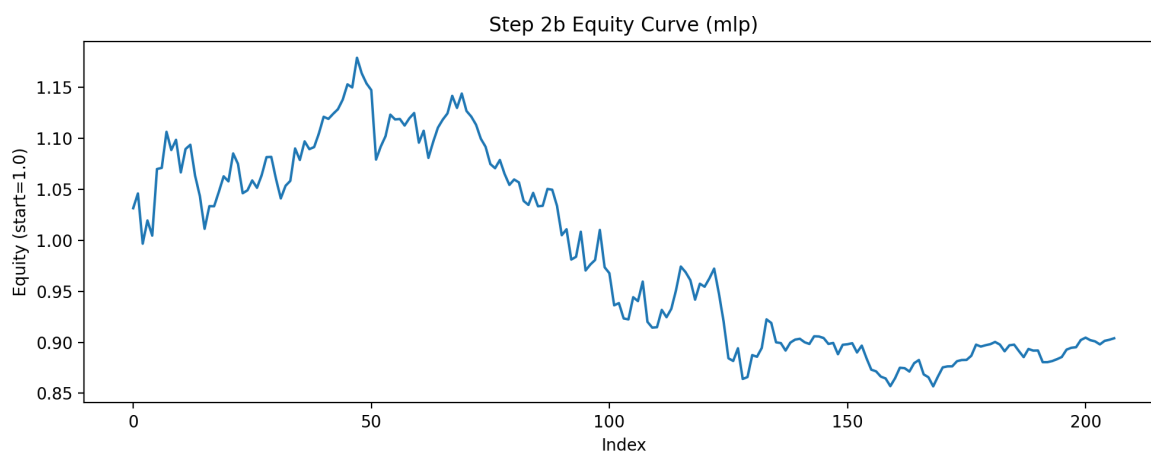


Figure 10: Step 2b aggregated walk-forward equity curve (MLP).

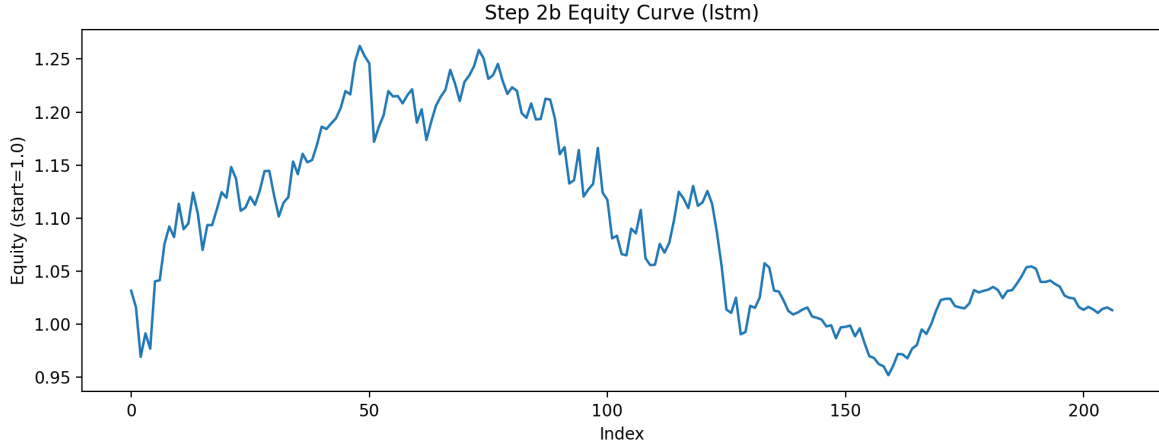


Figure 11: Step 2b aggregated walk-forward equity curve (LSTM).

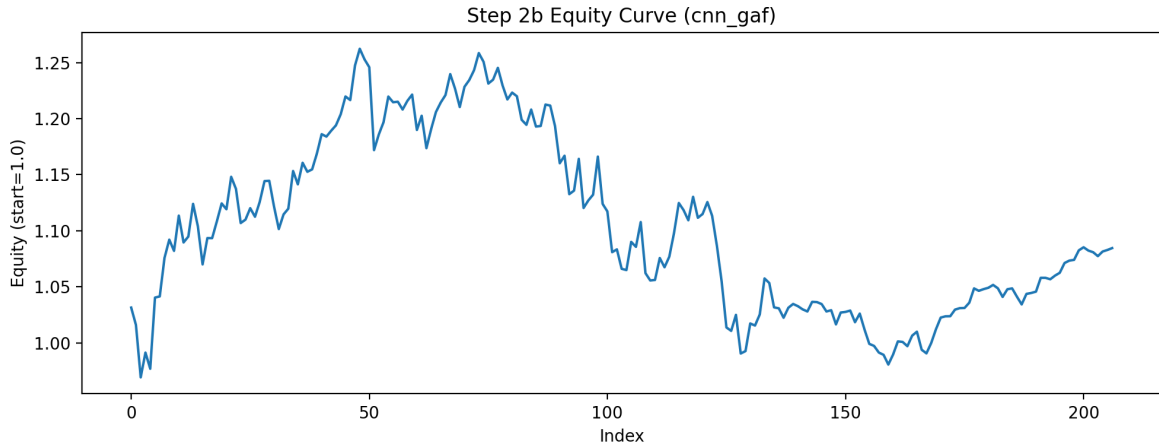


Figure 12: Step 2b aggregated walk-forward equity curve (CNN on GAF).

3.3 Step 2c. Comparison to Step 1

Walk-forward backtests (non-anchored) are evaluated under two regimes: 500/500 (Step 2a) and 500/100 (Step 2b). Step 2 keeps leaky preprocessing (global scaling), which can increase the probability of selecting or reporting over-optimistic backtests (López de Prado; Bailey et al.).

Model	Sharpe(1)	Sharpe(2a)	Sharpe(2b)	TotalRet(1)	TotalRet(2a)	TotalRet(2b)
mlp	1.271	0.746	0.547	0.526	0.567	0.090
lstm	1.417	1.174	0.148	0.606	1.098	0.005
cnn_gaf	1.315	1.049	0.370	0.548	0.927	0.051

3.4 Step 2d. Comparison between Step 2a and Step 2b

Step 2a (500/500) uses longer test windows and fewer model re-fits than Step 2b (500/100), which uses shorter test windows and more frequent re-training. Under leaky preprocessing, repeated

short-horizon evaluation can amplify backtest over-optimism (López de Prado; Bailey et al.).

Model	Acc(2a)	Acc(2b)	Δ	AUC(2a)	AUC(2b)	Δ
mlp	0.536	0.546	+0.010	0.511	0.503	-0.008
lstm	0.543	0.536	-0.006	0.506	0.485	-0.021
cnn_gaf	0.545	0.556	+0.011	0.504	0.522	+0.019

Model	Sharpe(2a)	Sharpe(2b)	Δ	TotalRet(2a)	TotalRet(2b)	Δ
mlp	0.746	0.547	-0.200	0.567	0.090	-0.477
lstm	1.174	0.148	-1.026	1.098	0.005	-1.093
cnn_gaf	1.049	0.370	-0.679	0.927	0.051	-0.875

If Step 2 performance is materially higher than Step 3 after leakage reduction, that pattern is consistent with backtest overfitting driven by leakage.

4 Step 3 (Walk-forward backtests with leakage reduced)

4.1 Step 3a. Leakage-reduction method

We reduce leakage by enforcing time-series-safe preprocessing and split hygiene:

- Fit feature scalers inside each fold using the training window only, then transform training and test separately.
- Purge: drop the last h observations of each training window (where h is the label horizon) so labels cannot overlap the test period.
- Embargo for lookback: skip the first 32 observations of each test window so test samples cannot reuse lookback information from the training boundary.
- For GAF images, normalize returns using training-only min/max per fold (no global min/max).

These practices follow time-series cross-validation guidance for financial ML (purging/embargo and leakage control) (López de Prado).

4.2 Step 3b. Walk-forward with 500/500 splits (leakage reduced)

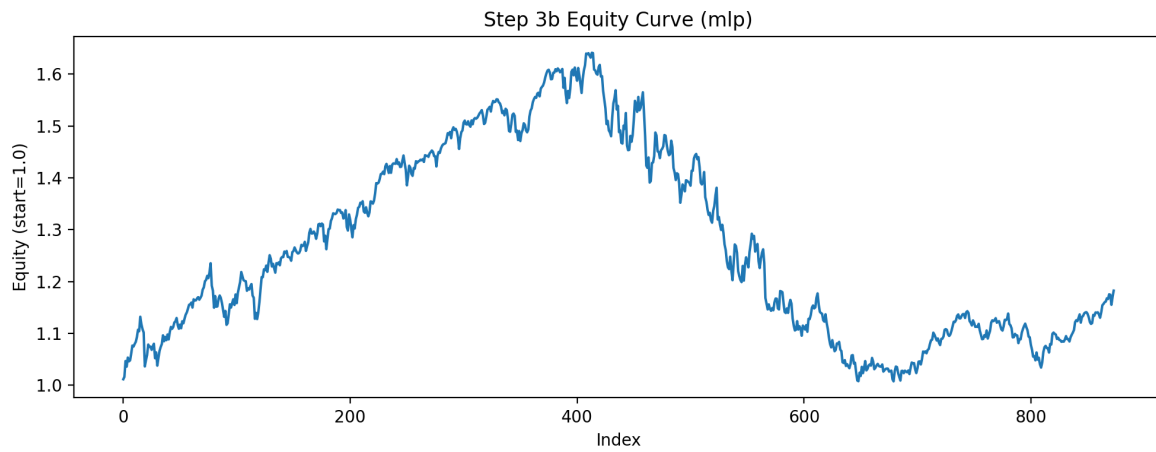


Figure 13: Step 3b aggregated walk-forward equity curve (MLP).

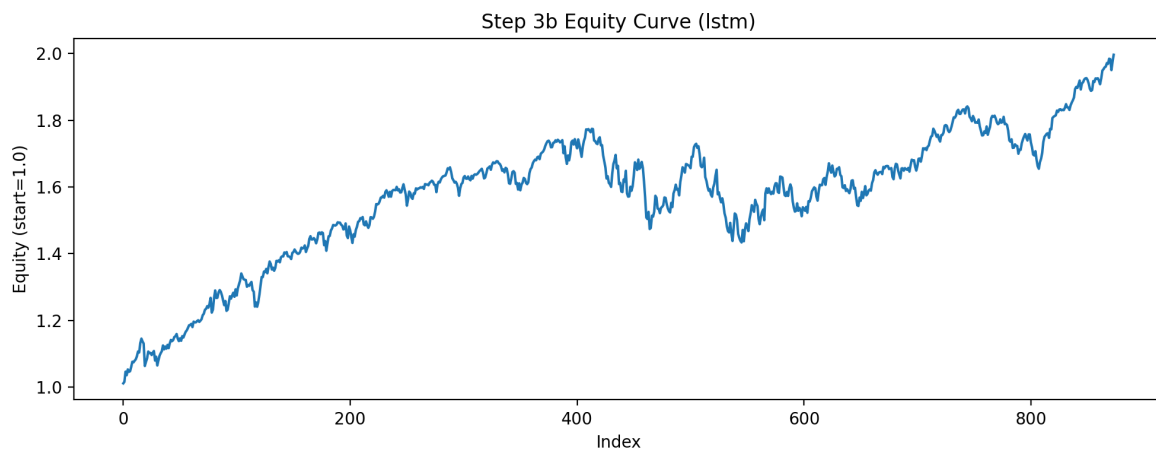


Figure 14: Step 3b aggregated walk-forward equity curve (LSTM).

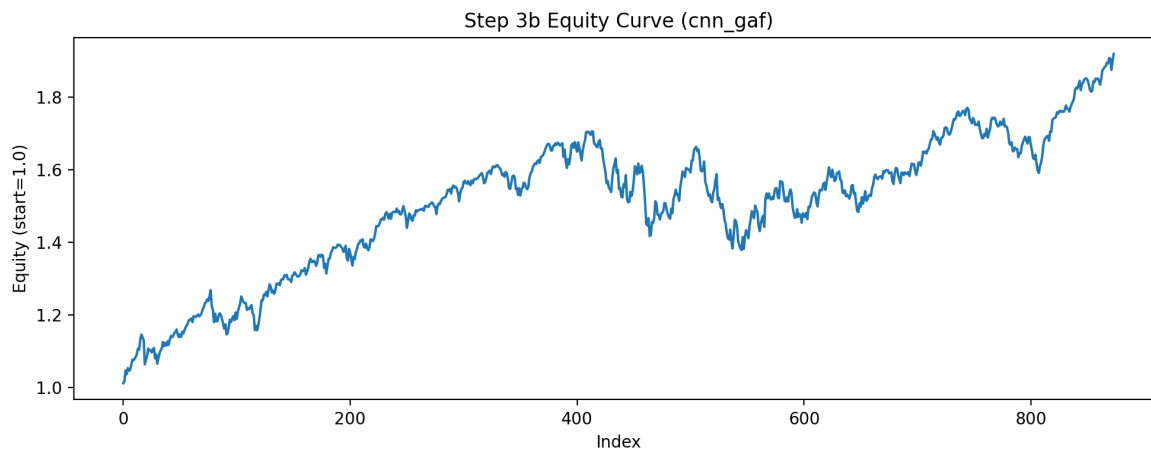


Figure 15: Step 3b aggregated walk-forward equity curve (CNN on GAF).

4.3 Step 3c. Walk-forward with 500/100 splits (leakage reduced)

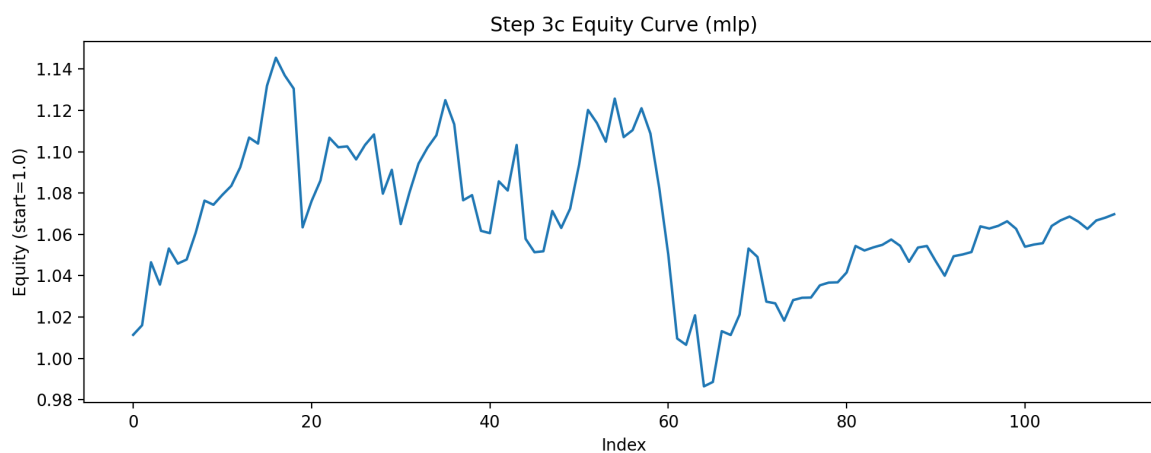


Figure 16: Step 3c aggregated walk-forward equity curve (MLP).

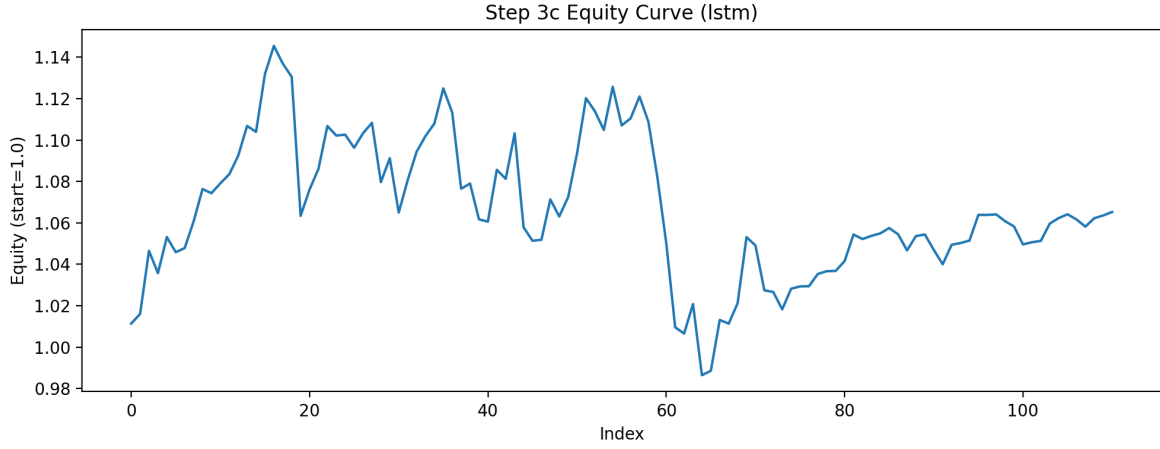


Figure 17: Step 3c aggregated walk-forward equity curve (LSTM).

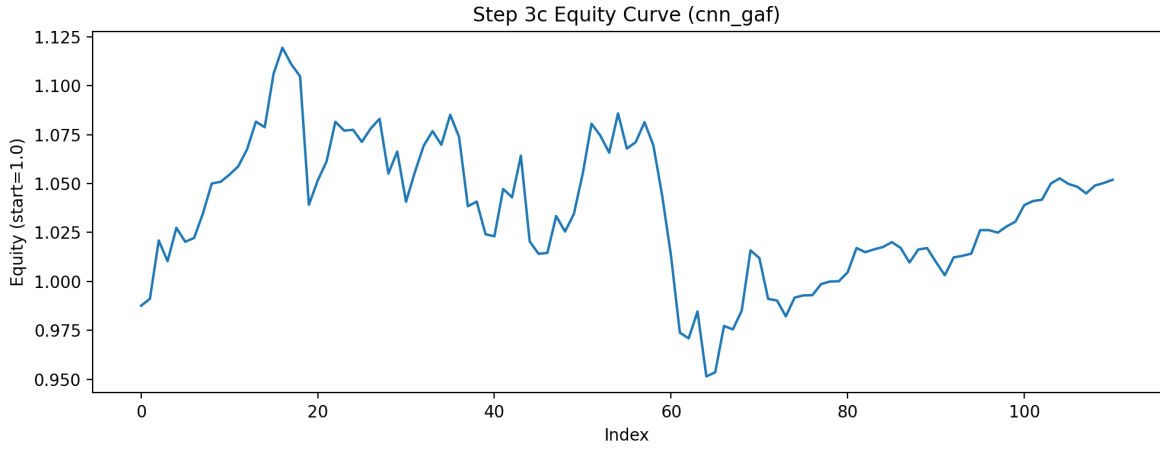


Figure 18: Step 3c aggregated walk-forward equity curve (CNN on GAF).

4.4 Step 3d. Comparison of Step 3b and Step 3c; has overfitting disappeared?

To assess whether Step 2 benefited from leakage-driven overfitting, we compare Step 2 (leaky) to Step 3 (leakage reduced) under matched walk-forward settings.

Model	Sharpe(2a)	Sharpe(3b)	Δ	Sharpe(2b)	Sharpe(3c)	Δ
mlp	0.746	0.367	-0.379	0.547	0.779	+0.232
lstm	1.174	1.253	+0.079	0.148	0.738	+0.589
cnn_gaf	1.049	1.186	+0.137	0.370	0.613	+0.244

Model	TotalRet(2a)	TotalRet(3b)	Δ	TotalRet(2b)	TotalRet(3c)	Δ
mlp	0.567	0.182	-0.384	0.090	0.070	-0.020
lstm	1.098	0.996	-0.102	0.005	0.065	+0.060
cnn_gaf	0.927	0.919	-0.008	0.051	0.052	+0.001

If Sharpe and total return systematically drop when moving from Step 2 to Step 3 (especially under matched windows), this is consistent with leakage-driven over-optimism in Step 2 (López de Prado; Bailey et al.).

5 Conclusion and Practical Takeaways

- Walk-forward evaluation is materially harder than a single split because it exposes the model to regime changes and repeated re-training, which typically reduces stability.
- Comparing Step 2 (leaky scaling) to Step 3 (fold-specific scaling + purging + embargo) shows how leakage can inflate backtest performance; the Step 3 results are the preferred estimate of deployable performance.
- Under leakage-reduced testing, the strongest configuration in this run is **lstm** in **Step 3b** with Sharpe=1.253, TotalRet=0.996, MaxDD=-0.192.
- Practical deployment should add position sizing, transaction-cost stress tests, and a strict research protocol (no tuning on test folds) to reduce the probability of backtest overfitting (López de Prado; Bailey et al.).
- In broader asset-pricing contexts, neural networks and other ML methods can add economic value, but only under careful out-of-sample evaluation (Gu, Kelly, and Xiu).
- If performance is only marginal after leakage reduction, prioritize robustness (simpler models, fewer degrees of freedom, and longer test windows) over headline returns.

6 Works Cited (MLA)

Bailey, David H., Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. “The Probability of Backtest Overfitting.” *The Journal of Computational Finance*, vol. 20, no. 4, 2017, pp. 39–69. doi:10.21314/JCF.2016.322.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu. “Empirical Asset Pricing via Machine Learning.” *The Review of Financial Studies*, vol. 33, no. 5, 2020, pp. 2223–2273. doi:10.1093/rfs/hhaa009.

Hochreiter, Sepp, and Jürgen Schmidhuber. “Long Short-Term Memory.” *Neural Computation*, vol. 9, no. 8, 1997, pp. 1735–1780. doi:10.1162/neco.1997.9.8.1735.

LeCun, Yann, Léon Bottou, Yoshua Bengio, and Patrick Haffner. “Gradient-Based Learning Applied to Document Recognition.” *Proceedings of the IEEE*, vol. 86, no. 11, 1998, pp. 2278–2324. doi:10.1109/5.726791.

López de Prado, Marcos. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

Wang, Zhiguang, and Tim Oates. “Imaging Time-Series to Improve Classification and Imputation.” *arXiv*, 1 June 2015, arXiv:1506.00327. doi:10.48550/arXiv.1506.00327.

WorldQuant University. *MScFE 642 Deep Learning for Finance: Group Work Project #3*. WorldQuant University, 2022.